Electronic voice-output communication aids for temporarily nonspeaking patients in a medical intensive care unit: A feasibility study

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BACKGROUND: The inability to speak during mechanical ventilation is recognized as a terrifying and isolating experience that is related to feelings of panic, insecurity, anger, worry, fear, sleep disturbances, and stress among critically ill patients. Alternative methods of communicating with temporarily nonspeaking patients in the intensive care unit (ICU) have received little study. Although electronic voice output communication aids (VOCAs) are available for disabled children and adults, the effectiveness of VOCA systems with adult medical ICU patients who may have multisystem illness, prolonged intubation, and longer ICU stays has not been explored.

OBJECTIVES: The purpose of this pilot study was to describe (1) the characteristics of intubated MICU patients who use VOCAs, (2) the usage patterns (message categories, frequency, assistance required), (3) communication quality (ease, user satisfaction), and (4) barriers to communication with VOCAs.

METHODS: This pilot study used participant observation, semi-structured interviews, questionnaires, and clinical record review in a complementary design to obtain data on communication events and VOCA use with 11 critically ill adults.

RESULTS: Study participants, 45.5 ± 16.0 years of age with 13 ± 1.9 years of education and moderately severe illness (APACHE III = 27.5 ± 16.1), used the VOCA for 5.7 ± 4.6 days. Ease of Communication Scale measurements showed significantly less difficulty with communication after device use ($t > 2.62$; $P = .047$). Almost half ($n = 5$) of the participants demonstrated some independent use of the device. VOCAS were used in one quarter of observed communication events. Patients used VOCAs most often to communicate with family visitors and initiated communication interactions more often when VOCAs were used than when communicating by other nonvocal methods. Poor device positioning, deterioration in patient condition, staff time constraints, staff unfamiliarity with device, and complex message screens were primary barriers to VOCA use.

CONCLUSIONS: This study showed that use of VOCAs is possible with selected critically ill adults and may contribute to greater ease of communication during respiratory tract intubation particularly with family members. Further clinical research using control or comparison groups is needed. (Heart Lung® 2004;33:92-101.)

INTRODUCTION

The inability to speak during mechanical ventilation is recognized as a terrifying and isolating experience for many intensive care unit (ICU) patients. Studies of the experiences and stressful events among ICU patients show significant relationships between the inability to talk and feelings of panic, insecurity, anger, worry, fear, sleep disturbances, and perceptions of stress.[1-4] Alternative methods of communicating with temporarily nonspeaking patients in the ICU have received little study.[5,6] This article describes the usage patterns, user perceptions, and characteristics of 11 intubated medical ICU patients who used electronic voice output communication aids (VOCAs).

BACKGROUND

Prior studies demonstrated that most nurse-patient communication interactions in the ICU are

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brief, consisting of task or procedure-oriented information, commands, or reassurances.[7-10] Nurse-patient communication interactions are most often initiated and controlled by the nurse and are influenced by the patient's severity of illness, level of consciousness, and responsiveness.[7-11]

There are a few preliminary studies and some anecdotal reports of the effectiveness of Augmentative and Alternative Communication (AAC) systems in the ICU. AAC refers to all communication methods that supplement or augment natural speech.[12] Dowden and colleagues[13,14] described success of AAC strategies for 50 temporarily nonspeaking ICU patients who received AAC consultation during a 2-year period in which success was defined as patient use of the recommended system for more than 1 day. Although prosthetic oral approaches (eg, electrolarynx, tracheostomy speaking valves) were the most frequently recommended techniques, patients demonstrated higher usage rates for direct selection systems (eye gaze systems; word, picture or alphabet boards) and natural speaking approaches (writing, mouthing words, gesture). The most common reasons for intervention failure were decreasing cognition (51%), patient's rejection (27%), and decreasing motor control (20%).[14]

A picture board intervention implemented with temporarily intubated cardiothoracic surgical patients improved patient satisfaction with communication of basic needs during the post-surgical period.[15] Alphabet and picture boards were preferred over electronic devices by 4 of the 5 ICU survivors interviewed about their experiences with AAC in ICU.[16]

VOCAs are a subset of AAC devices that produce prerecorded, digitized voice messages (recorded speech) or synthesized speech (computer-generated voice) when the communicator accesses specific locations on a dynamic display screen or membrane keyboard. Most electronic VOCAs can be pre-programmed with situationally-relevant whole messages, such as "I'm having pain," that are accessed via one location on the device display. Pre-programmed messages on additional "levels" can be added for elaboration. For example, a numerical scale representing degree of pain can be activated after a simple pain message is communicated. Some devices allow the individual to spell new messages via multiple keystrokes.

Costello[17] described VOCA use with 43 patients, 2.8 to 44 years of age, at Children's Hospital in Boston. Before surgery, patients and family members selected and recorded 30 to 40 vocabulary items that were then categorized under topic cues, such as words or icons. Patients and families expressed general satisfaction with the communication intervention and patients did not report exhaustion, isolation, and/or fear related to the inability to speak in the ICU.[17] However, no measures of communication ease, frequency, quality, or success were used. There have been no published studies of the effectiveness of electronic VOCA systems with adult medical ICU patients who may have multi-system illness, prolonged intubation, and longer ICU stays.[6] The purpose of this study was to describe the usage patterns, user perceptions, and characteristics of intubated MICU patients who used VOCAs. The following research questions were addressed:

(1) What are the characteristics (age, severity of illness, technologic intensity, medical diagnosis, cognition, use of hearing and visual aids, upper extremity function, educational level, prior computer use, days ventilated, sedation/analgesia use) of patients who use a VOCA?
(2) What are the usage patterns (message categories, frequency, assistance required) of critically ill patients who use the VOCA to communicate with family members, and clinicians?
(3) What is the quality (ease of communication, user satisfaction) of patient-nurse, patient-family communication when a VOCA is used?
(4) What are the barriers to use of the VOCA?

METHODS

This study combined quantitative and qualitative approaches in a complementary design.[18] Participant observation, semi-structured interviews, questionnaires, and clinical record review were used to obtain data on VOCA use with 11 critically ill adults.

Setting

The study was conducted in a 20-bed medical ICU of an urban, university-affiliated, tertiary care hospital following review and approval by the University of Pittsburgh Institutional Review Board. Before this project, the MICU had not been exposed to AAC technology for nonspeaking patients other than electrolarynx and one-way tracheostomy speaking valves.

SAMPLE

Sixteen adult patients (primary participants) were selected from the MICU using the following criteria: (1) intubated, (2) responsive to verbal stimuli, (3) able to follow simple commands, (4) able to

Table I
VOCA device set-up

Features used	MessageMate	DynaMyte
Number of icons/messages on display	36	6-12/page
Spell?	No	Yes
Number of words in output message	1-7	1-10
Direct select (point)	Yes	Yes
Auditory scan/switch	No	Yes, Case 8
Visual scan/switch	No	Yes, Case 11
Battery power	Yes	No*
Dimensions (in)	$11.75 \times 4 \times 1.25$	$8 \times 7 \times 2$
Weight (lb)	1.66	3.20
Memory Card (message count)	No	Yes†

*Although equipped with a 6- to 7-hour rechargeable battery, the device was connected to a battery charger and power source continuously for this study.
†These data were of limited use. Power source interruptions deleted message counts for several cases. The unit tallied alphabet letters rather than full messages when the spell feature was used.

understand English, and (5) able to complete 6 of 8 Initial Cognitive-Linguistic Screening Tasks.[13] Cognitive Linguistic Screening Tasks is a clinical tool used to assess attention (Look at me.), orientation (Is Ronald Reagan the current president?), and the ability to follow directions (Close your eyes.) using yes/no questions and simple tasks.[13] Patients were purposefully selected according to qualitative exploratory sampling methodologies[18] for variation in diagnosis, length of mechanical ventilation, age, gender, education, prior history of mechanical ventilation, and prior history of computer use. Five MICU patients or their surrogates refused invitation to participate. The final study sample was 11 patients. Family members and critical care clinicians (nurses, physicians, speech-language therapists, respiratory therapists) were secondary participants.

Procedure

Patient participants were identified by one of the investigators (MBH or TR) during ICU rounds or by critical care nurse referral. After receiving informed consent from patient and/or family surrogate, the researchers collaborated with the patient's nurse to determine the appropriate time for patient VOCA instruction. Study participants were followed until extubation or hospital discharge, whichever occurred first. Investigators (MBH, TR) carried pagers, rotating "on call" to solve problems or answer questions about the device.

Equipment. Two VOCAs were used in this study. The MessageMate (Words+, Inc., Lancaster, Calif) is a battery-powered, small message capacity, multilevel recorder with digitized speech output. Patients touch a 1-inch by 1-inch word-picture icon on a keypad to activate a prerecorded message in the patient's or another person's voice. In contrast, the DynaMyte (Dynavox Systems, Inc., Pittsburgh, Pa) is a large message capacity, dynamic display device with synthesized (computer-generated) speech output. Touching a labeled picture on a dynamic display screen activates a voice message. Accessing certain symbols can "link" the communicator to related messages on a different "level," or screen. Patients can also spell new messages by using a touch screen keyboard. Both devices can be individualized by changing messages, icons/symbols, and the display to meet a patient's unique communication needs. Table 1 contrasts device features used for both systems in this study.

The DynaMyte device with VitalVoice software (Dynavox Systems, Inc., 1999) was used with the first 8 study participants. The MessageMate, set up in a 36-message, single level format using a research assistant's voice, was added to this study after new literature documented patient choice of this type of device in a surgical ICU setting.[17] Therefore, the next 2 participants received the MessageMate. The DynaMyte was selected for the final participant based on his needs for a special access (visual scanning) selection method. The small number of observed communication events with the MessageMate precluded comparisons between devices.

Vocabulary/message selection. Message displays and message selections were individualized on the basis of input from the patient or family. Patients were shown "standard" messages derived from the literature[3,7-10,17] and queried about possible additions or deletions, such as requests for eyeglasses, prayer, or music. "Standard" vocabulary/message options evolved during the course of this study as we received patient feedback on usefulness of particular items. Messages common to all cases included pain, shortness of breath, request for suctioning, help, hot/cold, home, family, anxiety and/or worry.

Training. Initial patient instruction sessions varied in length, but generally lasted 20 minutes and were discontinued if the patient appeared fatigued, in pain, unable to comprehend, or requested to stop. Additional instruction was conducted as needed throughout the period of device use. Nurses attended a 15-minute educational program on the study purpose and operation of the communication device. An instructional manual was available outside of the patient's room for review.

Patient characteristics. Demographic and clinical data were obtained by chart review and patient or family report. Acute Physiology and Chronic Health Evaluation (APACHE III) scores measured severity of illness on the day of study enrollment. The APACHE III has the advantage of being well accepted, and widely used.[19,20] Therapeutic Index of Severity Score (TISS)[21] is a quantification of 76 therapeutic interventions used to measure technologic intensity at time of entry into the study ($r = 0.96$).[22] The Glasgow Coma Scale (GCS) was used as a gross measure of cognition (ie, arousal or wakefulness) on entry into the study and at the time of each observed communication event, with appropriate modifications to verbal score for intubated patients.[23] Specifically, patients received full verbal score points if they were able to consistently communicate (yes/no, or writing, or gesture) via nonverbal means. Interrater reliability for the APACHE III, TISS, and GCS was maintained at >0.90 by testing a randomly selected 10% sample of case data using item agreement tallies.

Outcome measures. To assess changes in patient perception of communication difficulty, participants completed the revised Ease of Communication Scale (ECS),[1] before introduction of the VOCA and after the VOCA had been used. The researcher read the 10 Likert-type statements about perceived communication difficulty to patients who referred to a card with response selections (0 = not hard at all, 1 = a little hard, 2 = somewhat hard, 3 = quite hard, 4 = extremely hard) printed in large font. The original 6-item ECS showed good internal validity ($\alpha = 0.88$).[1] Internal consistency for the revised ECS was established in this study at $\alpha = 0.93$.

Study participants were observed for at least 20 minutes each day for communication interactions. The investigator-developed Observation of Communication Event Record was used to document characteristics of observed communication interactions between patient and communication partner (ie, nurses or family visitors) such as initiation of communication, position of device, methods of communication, message content, frequency of VOCA use, and assistance required by the communication partner. Patients, clinicians, and family members were asked informally about their experience with the device, including validation of message content heard by observers. Clinical records were also reviewed for documentation of nonvocal communication method, content, and VOCA use. Observational interviews and clinical record review were documented in field notes. The Communication Methods Checklist,[1] a summation of communication methods by report (from patient, family, nurse) and direct observation while the patient is unable to speak, was reviewed and updated daily.

Finally, 14 formal interviews were conducted with 8 patients, 3 family members, and 3 clinicians (2 registered nurses, 1 respiratory therapist) to ascertain satisfaction with the device, barriers to use, and common messages from the perspective of the study participants and their communication partners. The VOCAs were used during the interview to question participants about each screen and message option. Interviews were audiotaped whenever possible, transcribed verbatim, reviewed for accuracy and corrected by the interviewer.

Data analysis

Quantitative data were analyzed using descriptive statistics (mean, SD, frequency) and pattern identification via data matrices.[24] Characteristics of communication interactions (ie, communication method(s), number of communication partners, role of communication partner, position of partner, patient, and device, initiator of message, message validation, sedation/analgesia, physical restraint use) were coded from the Observation of Communication Event Record and tabulated.

Qualitative field note data were analyzed by simple coding and categorization of the following: (1) communication method, (2) barriers to use of VOCAs or other nonverbal communication technique,

Table II
Patient characteristics (n = 11)

Case	Diagnosis	Device days	Vent days	Days intubated prior to device	APACHE III
1	Pneumonia	2	7	4	24
2	Sepsis	11	22	11	51
3	ARDS	1	3	1	10
4	Pneumonia	5	25	27	30
5	Pneumonia	5	12	6	54
6	Respiratory failure	2	10	8	47
7	Aspiration pneumonia	3	10	9	22
8	MS	14	18	4	10
9	Respiratory failure	1	0.5	0.5	16
10	Subglottic stenosis	11	19	16	24
11	MVA/SCI	8	44	46	15
Mean		5.73	15.50	12.05	27.55
SD		4.58	12.20	13.57	16.11

ARDS, Adult respiratory distress syndrome; *MS*, multiple sclerosis; *SCI*, spinal cord injury; *Device days*, number of days patient with VOCA device; *Vent days*, time period in days of mechanical ventilation from initiation to either extubation, successful speaking device use or death; *APACHE III*, APACHE III score on enrollment day; *ECS*, Ease of Communication scale: possible score 0-40 (40=highest difficulty); *Usage category*, 1-little to no use; 2-occasional use with cueing; 3-some independent use; 4-dominant form of communication; *Airway*, type (ETT-endotracheal tube, Trach-tracheostomy); *Sedation/analgesia*, intermittent anxiolytic and/or analgesia use within 6 hours of communication event; *Propofol infusion (at enrollment and during device use/device days); *Visual correction*, Y-correction necessary; *correction available; #-blind.

(3) facilitators of use of the device or nonverbal communication technique, and (4) content of patient communication.[24,25] Data were coded independently by 2 reviewers with final codes designated by the principal investigator (MBH). The following categories of VOCA usage were developed from the observation data, patient, family, and nurse report and documentation in the clinical record: (1) little to no use, (2) occasional use with cueing, (3) some independent use, and (4) dominant form of communication. Two independent raters categorized VOCA usage by reviewing data files for each study participant according to these categories. Because cognitive and motor abilities fluctuate so quickly and dramatically in the medical ICU, we categorized patient device use according to their "best use." Discrepancies were resolved by consensus agreement after review of the data and category definitions.

RESULTS

Patient characteristics

Eleven MICU patients, ranging in age from 24 to 72 years (45.5 ± 16.0) were recruited to participate in the study. Primary admitting respiratory diagnoses included pneumonia, acute respiratory distress syndrome, lung cancer, chronic obstructive pulmonary disorder, and subglottic stenosis. Demographic and clinical characteristics for study participants are described in Table 2. Few (n = 3) participants could write legibly. Four patients demonstrated difficulty or were unable to complete the motor screening tasks at study enrollment. One of these patients (case 8) was blind and quadriplegic with spasticity and limited gross motor ability of bilateral upper extremities. A second quadriplegic patient (case 11) also had limited gross motor ability of the upper extremities. The VOCA was adapted with scanning devices and switch techniques for these patients. Others (cases 2 and 4) had weak fine motor coordination and experienced difficulty in positioning and reaching the device. They used thumbs, fingernails, pens, and extension wands to touch activate the VOCA display screen.

Seven patients received narcotic analgesics (n = 14) and/or anxiolytics or sedation (n = 25) within 6 hours of study observation periods. Specifically, narcotic analgesia and/or anxiolytics and sedatives were present in 39% (n = 16) of observations in which communication events occurred (n = 41).

TISS score	ECS pre score	ECS post score	Usage category	Airway	Sedation/ analgesia	Visual correction
32	N/A	10	2	ETT	Y*	*Y
42	40	27	2	Trach	Y	*Y
38	38	17	3	ETT	Y*	*Y
25	23	17	3	Trach	Y	*Y
29	40	N/A	3	ETT	Y*	*Y
32	39	23	2	ETT	N	*Y
30	N/A	N/A	2	Trach	N	N
42	N/A	N/A	1	ETT	Y	Y#
35	8	7	3	ETT	N	Y
22	23	24	1	ETT	N	N
33	38	N/A	1	Trach	Y	*Y
32.7	31.13	17.86	2.09			
6.3	11.84	7.40	0.83			

Physical restraint use was rare (twice in 49 observations). Although not a condition for enrollment or observation, all patients had a high GCS (14-15) at communication observations.

VOCA Usage Patterns

Frequency. Forty-nine observations were conducted and recorded. Actual patient communication events were documented in 83.7% (41/49) of the observation periods. VOCAs were used in one quarter (11/41) of the observed communication events. However, more than 1 method of communication was employed during most interactions (29/41, 70.7%). Specifically, in constructing messages with the VOCAs, patients used gesture ($n = 6$), mouthing words ($n = 6$), head nods ($n = 4$), and writing ($n = 1$). No participant used the VOCA as the dominant method of communication, however, most patients used the VOCA independently or with cueing (see Table 2). On interview, patients reported using VOCAs more often with their family visitors than with nurses.

Assistance required. Patients often stabilized the device by resting the palm of the hand on top of device and using the thumb to perform key selections. Most VOCA-constructed messages (10/11) required some assistance (eg, repositioning, cueing, review of message options, or message location) and/or validation (eg, repeating message or asking questions) from communication partners. Patients initiated communication more often in observed communication events involving VOCAs (36.4%) than in communication events *not* using VOCAs (16.7%) (Fig 1).

Message content. VOCA-constructed messages were varied, ranging from simple commands (eg, "help!") to typed questions (eg, "Why am I on the ventilator?"). The primary content of observed communication messages were, "I love you" ($n = 9$), questions about home/family, extubation, and restraint release ($n = 5$), worry, fear, or anxiety ($n = 4$), pain ($n = 3$), and comfort needs ($n = 3$). The father of a 20-year-old college student who survived acute respiratory distress syndrome described the impact of one such message:

"You have a son and his parents and they walk into the room and he was hunting for and pushed 'I love you.'. . .and look how it affected [his mother]. It was almost like his own voice telling her that."

Communication quality

Ease of communication. Nonparametric within-case tests applied to the 6 patients able to complete

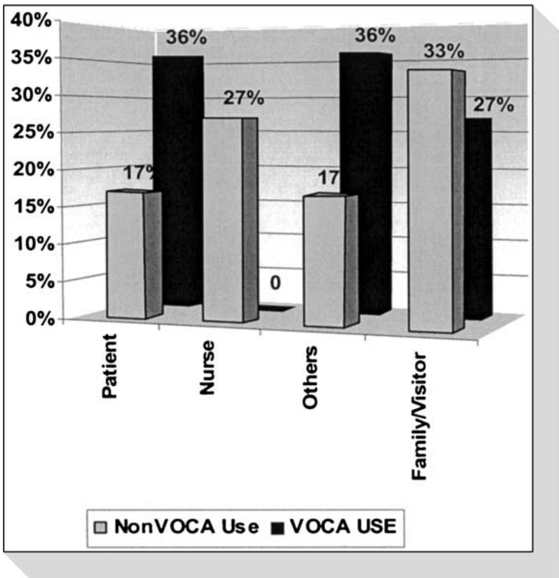

Fig 1 Initiation of observed communication (n = 41).

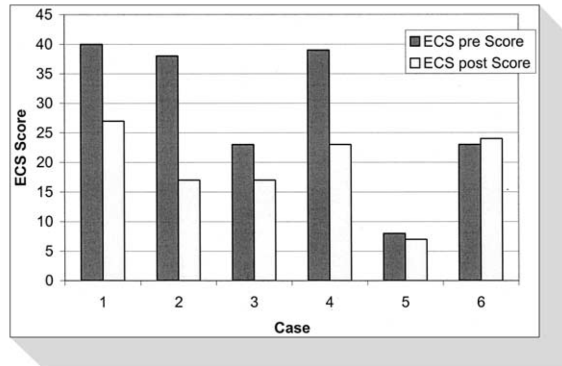

Fig 2 Pre- and post-intervention ease of communication scores.

the ECS at pre-intervention and post-intervention time points showed significantly less difficulty with communication after device use ($t > 2.62$; $P = .047$) (Fig 2). Post-intervention ECS scores (17.8 ± 7.4) were notably lower than pre-intervention ECS scores (31.1 ± 11.8), with higher scores indicating greater communication difficulty (Table 2). Case 10 demonstrated a slight increase in perceived communication difficulty after device use that may be related to length of time intubated (nonspeaking) in a non-study ICU before device use (Table 2) or psychosocial factors, such as depression, not measured in this study.

Satisfaction. Most feedback from patients and families was positive for both devices. A patient commented, "That" (pointing to the VOCA) "was a good thing there; it really helped me." Overall, family members found the patient easier to understand when the VOCA was used. "It was easier to understand what she wanted. I can't read sign language. . . I had to guess. . . I'm not a good guesser." Nevertheless, patients and families had several suggestions to improve the design and application of VOCAs in ICU, such as larger screens, greater or more reliable touch sensitivity, easier keyboard access (DynaMyte), and backlighting (MessageMate).

Clinicians expressed enthusiasm for the VOCA concept tempered with realistic concern about device complexity and patient abilities.

"I liked it. . . I think it's more complete and decisive. You know there's no guess work, 'cause if they accidentally pressed the wrong button, they would shake their heads vigorously 'no' or roll their eyes. For the most part, they got their point across more completely." —ICU *nurse*

"When they had [VOCAs] and they got the hang of this, they used it almost as a sole means of communication. They like [VOCAs] and they like the fact that people tend to respond to voice. And this was their voice. . . people respond to voice." —ICU *nurse*

Barriers to VOCA use

Poor device positioning, deterioration or fluctuation in patient condition (motor and/or cognitive function), staff time constraints, staff lack of familiarity with the device, and device complexity (multilevel message screens) were the primary barriers to VOCA use.

Poor device positioning. After moving the device to perform care, nurses and therapists often failed to replace or reposition the device so that the screen was easily visible or accessible to the patient. In some instances, the swivel arms supporting the device did not have the necessary length for optimum device positioning. In addition, screws on swivel arms required frequent tightening to prevent slippage of the device. Clinicians described the method of attachment as "cumbersome" and complained that "it gets in the way."

Deterioration/fluctuation in patient condition. A respiratory therapist commented, "The worst part of [the device] was a lot of [patients]. . .didn't have the strength to do that." Indeed, these medically frail patients were easily fatigued and motor ability fluctuated with improvements or declines in patient condition. Digital clubbing made activation of dynamic display screen difficult for one patient. During interviews, patients reported transient blurred vision as another obstacle to device use.

Similarly, there were wide fluctuations in cognition and attention among these patients. It was common for ICU patients to be attentive for completion of the initial cognitive screening tasks required for study enrollment but to fall asleep during device instruction. Although all patients were able to successfully generate valid messages with the VOCAs at some point during the intervention, profound psychomotor retardation was noted during intravenous sedation or analgesia in some patients.

Device complexity. Critically ill patients forgot message options, message locations, and how to return to the main screen on the DynaMyte. Patients commonly typed, "suction," or, "I'm thirsty," on the DynaMyte despite the availability of single-key options for those messages.

Staff time constraints and lack of familiarity with device. Most nurses continued to rely on yes/no questions, head nods, and lip reading to communicate with nonspeaking patients.

"It was easier for me to talk with him, and not have to pull out the device, because time is precious around here. . . Where he could get his point across to me with lip-talking, it seemed to lessen the time I needed to pull out the device, get it over near his bed, have it within reach so he could point to the different pictures on screen." —ICU *nurse*

Several nurses commented that they did not know what the device was or how to use it. This may have been caused, in part, by high staff turnover and the use of temporary nurse staffing during the 7-month study time period.

LIMITATIONS

A small, purposefully selected sample of patients was used in this study. Selection criteria (performance of Cognitive-Linguistic Screening Tasks) excluded the most seriously ill patients, agitated patients, and those with acute brain injury. In addition, MICU nurses were reluctant to recommend study participation for restrained patients because VOCA use would require upper extremity restraint removal, which further excluded potential study candidates.

Given the exploratory nature of the study, patient selection, device characteristics, instruction and training of patients and communication partners (nurses, family members), and vocabulary and message selection were not systematically manipulated, nor was a speech-language pathologist used to assist with matching device features to patient capabilities. Interestingly, the observers documented that most of the patients in this study had the capability to use these VOCAs, even though device features had not been matched to patient's abilities. Only 2 patients with significant neuromotor disability did not appear to have the capability to use the systems as designed. Future studies should accommodate a variety of motor, visual, and cognitive disabilities via device options and adaptations.

Measurement of the success or effectiveness of AAC interventions with critically ill patients has not been defined in the literature. This pilot study provided experience with research methods that would be useful in a more controlled intervention study. Though the ECS is a valid measure of communication difficulty, not all patients who met study entry criteria were able to complete the survey. A single-item analog scale of communication difficulty may be more feasible in this population.

Finally, the observational data are limited by distractions in the ICU environment, such as high noise levels, isolation rooms, and multiple simultaneous interactions.[26] Patient, family, and clinician reports indicated greater use than we were able to directly observe. Therefore, these data present a conservative depiction of patient communication in general, and VOCA use in particular.

DISCUSSION

Patient characteristics

Adult patients of all ages, educational levels, and with moderate-to-low illness severity were able to use the VOCAs. The patients in this study had relatively high levels of cognitive function (GCS scores = 14-15). Neuromotor disability posed the greatest challenges to VOCA use in the critical care setting, as patients who were unable to successfully complete the Motor Screening Tasks used the VOCAs minimally.

Usage patterns

The success of a VOCA-based AAC intervention within a given ICU population is dependent on many factors, including the availability of AAC technology, patient capabilities, device complexity, and partner familiarity and training.[27-29] In addition, fluctuations in ICU patients' cognitive, motor, and medical status, often on a moment-by-moment basis, clearly influence ability to effectively use available AAC systems. Most importantly, 8 out of 11 (73%) patients in this study were able to use VOCAs with minimal assistance and instruction despite serious illness, poor positioning, sedation, narcotic analgesia, and no "feature match."

Most nonspeaking ICU patients will use more than 1 technique or strategy during their ICU stay.[13,14,16] In fact, multimodal or combination solutions and customized AAC techniques have become the standard of care in AAC practice.[30] Dowden et al[13,14] found natural speaking approaches (writing, mouthing words, gesture) had best success rates of all types of AAC methods recommended for ICU patients. Similarly, we observed and documented a high rate of use of these approaches alone and in combination with VOCAs. Costello reported pre-op selection preference for simplier multi-level recorder devices like MessageMate.[17]

The most striking finding in this study is that patients initiated more than one-third of the observed communication interactions involving VOCAs. This differs from previous studies of nurse-patient communication in which nurses were more often the initiators of communication.[7-11] Use of VOCAs may permit critically ill patients more control or equity in communication interactions. Patient reports that VOCAs were used most often with family members further indicate that VOCAs provide a different pattern of communication. This is particularly interesting in light of previous research in which mechanically ventilated patients reported greatest frustration and difficulty in communicating with family.[1] In this study, expressions of sentiments (eg, "I love you") and feelings (eg, worry, fear) were the most frequently observed messages, most often initiated by the patient, and directed to family members. If critically ill patients truly need to construct more complex messages with family members than with nurses as suggested by Menzel,[1] VOCAs may facilitate construction and delivery of those messages. The lack of observed VOCA communications initiated by nurses (0%) may be caused by insufficient education and training in use of VOCAs or to greater use of yes/no questions and anticipatory communications on the part of nurses.

Communication quality

Ease. Although some patients were unable to complete the revised ECS, data from the 6 patients with pre and post-intervention ECS scores showed notable reductions in perceived difficulty with communication after using the VOCA. Without a comparison group, we are unable to attribute the improved ease in communication to VOCA use. These patients may have perceived any attention to communication difficulties as helpful.

Barriers to VOCA use

The barriers to VOCA use identified in our study are congruent with the most common reasons for AAC intervention failure described by Dowden and colleagues.[14] The current research identified specific technical and social factors such as device positioning, design features, and staff time and preparation that expand our understanding of the dynamics involved in implementing AAC interventions in the ICU setting.

Critical illness affects short-term memory;[31-34] therefore, AAC device formats should be easily recognizable, or "transparent," and instructions may need to be reviewed frequently with ICU patients. Having forgotten message options and locations, study patients often expressed surprise when reviewing device features in the post-intervention interviews. Optimally, patient understanding and ability to use the device should be tested daily with changes in functional ability documented and necessary adjustments made to the AAC plan of care.

The observation of profound psychomotor retardation in patients receiving sedatives and narcotics indicates a need for measurement of psychomotor task performance time in future communication research with this population. In addition, this finding suggests that critically ill patients' communication abilities may easily be underestimated by busy critical care clinicians who are unable or unwilling to wait for slow message generation to be completed.

Device positioning and access for patients with limited head, neck and upper extremity movement was an additional challenge. The devices needed frequent repositioning and adjustment as the patient's position changed. For some patients, the activity involved in using the device (and in supplemental communication via mouthing words) stimulated coughing episodes. Though thumb control (mouse or switch) options seem like a good solution, the coordination, practice, and new learning required may be unrealistic for most critically ill patients.[32] On the basis of this research experience, the following optimal VOCA features are suggested for hospitalized patients: (1) easily programmable, (2) long battery life (>24 hours), (3) 2-3 basic message display formats that can be individualized and saved like a word-processing file, (4) few speaker holes or crevasses for cleaning, (5) able to withstand nonabrasive, antibacterial cleaner, (6) light weight, (7) large, backlit display screen ($\geqslant 8 \times 11$), and (8) high quality speech.

CONCLUSIONS

This pilot study showed that use of VOCAs is possible with selected critically ill adults and may contribute to greater ease of communication particularly with family members during respiratory tract intubation. Barriers to VOCA use may be addressed by design improvements, enhanced staff education, individualized assessment, and combination strategies. This study provided preliminary description of VOCA usage patterns and user satisfaction in a previously untested MICU population. A more controlled investigation comparing the communication patterns of randomly selected adults who use VOCAs to those who do not use VOCAs to communicate in the ICU is warranted.

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